Build a voice agent that builds travel agents

Three-hour masterclass · Rasa Mantle, Codex and the hosted OpenAI Agents API

FIRST CALL

Speak as a traveler

Ask a travel desk for a forecast and a currency conversion. Hear the reply and trace the numbers to their tools.

FINAL CALL

Speak as a developer

Commission a new travel desk. Review what the factory produced, run it and call it as a traveler.

Which forecast belongs to my flight?

“I fly Oslo to Bergen tonight. What should I pack, and what is $200 worth there?”

1

Ask first

Where will you land? At what local hour?

2

Evidence

A route, an arrival time, a forecast hour and a dated exchange rate.

Tomorrow the travel team asks for a change

“When two records match my flight number, ask which date I mean before giving me the weather.”

TODAY

One desk works

It can answer a straightforward question about one flight. A demo of that trip tells us little about the ambiguous case.

TOMORROW

A second desk must work too

How do we turn that request into a new Mantle skill and check that it asks before calling the weather tool?

How a request becomes a second agent

A software factory is a repeatable way to turn a request into checked software. Our factory produces a runnable Rasa Mantle project.

1 · Describe	A developer gives a brief. The builder proposes a specification and skill instructions.
2 · Review and build	Inspect the candidate, verify its saved bytes and approve it. The compiler combines the skill with the selected travel tools.
3 · Try the product	Train the new desk, call it and inspect its behavior on the original trip and the changed request.

What runs where?

GitHub Codespaces gives everyone the same browser workspace.

Rasa Mantle + Deepgram	Mantle runs both conversations and calls their tools. Deepgram turns speech into text and the reply back into audio.
Codex	At the workbench, you use Codex to author a travel agent and refine its skill.
Hosted OpenAI Agents API	The factory calls a hosted build session and downloads its candidate for review.
Travel data	Avinor supplies flight boards, Open-Meteo supplies forecasts and Frankfurter supplies exchange rates.

The route through the masterclass

The first checkpoint is a two-turn voice call by minute 30.

00–30 · Use the desk	Get live data. Ask two questions aloud and inspect their tool results.
30–75 · Build with Codex	Create the four-tool desk and refine how it handles ambiguity.
75–85 · Break	Save the run ID and your project directory.
85–135 · Operate the factory	Speak a build brief, review the candidate and start the new agent.
135–180 · Test and take home	Check the trip, exercise missing data and choose an independent extension.

Open the travel Codespace

Workshop README · Follow A–D, choose Rasa Travel Agent Factory and a four-core machine.

1

GitHub → Settings → Codespaces

RASA_LICENSE · OPENAI_API_KEY · DEEPGRAM_API_KEY for voice.

2

Check

Grant the workshop repo access. Open Terminal → New Terminal. Run bash desk doctor.

How much is $200 in pounds?

Copy the command from the README. Look at the direction of the currency pair.

```
bash desk data fx USD GBP 200

Example · 2 Nov 2026
200 × 0.74628 = 149.26 GBP
```

Read the receipt

Find converted and reference_date. Ask whether that rate is a guarantee from a card issuer.

Ask for the right weather hour

A forecast is indexed by airport and time. LHR is London Heathrow.

```
bash desk data weather LHR

Current forecast hour at the destination airport.
```

At landing

Pass an arrival timestamp with its offset. Report the forecast for that hour.

Call your first Mantle travel agent

Run the starter in Terminal 1. Leave that terminal running.

```
bash desk starter

bash desk inspect builds/starter
```

Terminal 2 → bash desk url

Start voice conversation, allow the microphone and ask about Heathrow weather. Listen, then ask for USD to GBP in the same call.

Where would you change this answer?

Open the starter project. Follow one question all the way to its source.

Routing → skill.md	When should Mantle select the travel skill? What needs clarification?
Execution → tools.py	Which provider does the tool call? Which data and timestamps enter ToolResult?

Write the brief for the full agent

One flight number matches two dates. Should the skill pick the first or ask which one the traveler means?

1

Input

Four capabilities: flight status, departures tonight, destination weather and FX.

2

Output

Codex authors agent.json. Read its routing description, instructions and selected tools.

Turn the candidate into a Mantle project

Read the candidate before approving it. Use the run ID from your own receipt.

```
bash desk approve runs/YOUR_RUN_ID

                    builds/codex-desk
```

What gets built

agent.yml + integrations.yml + a travel skill + registered Python tools.

What does “tonight” mean at Oslo?

Ask your full agent for OSL → BGO tonight. Compare its answer with the direct flight tool.

Local window	Remaining departures from 18:00 to midnight in Europe/Oslo.
Coverage	Matching records on the retrieved page. A schedule gives no seat inventory.

Save your place

Save your run ID and the name of the approved project.

1

Pause

Stop the traveler Inspector before starting Studio on the same port.

2

Resume

You can train an existing output again without generating another proposal.

Two Mantle agents have different customers

The traveler asks for information. The developer asks for a new traveler agent.

1

Developer → Studio

Mantle confirms one billed build, then calls the hosted builder.

2

Traveler → generated desk

The reviewed agent calls public providers when answering a trip question.

How do the bytes leave the hosted workspace?

A session says “ready” and returns only notes.txt. What can you actually build?

Create → finish	sessions.create starts the hosted environment. The final result supplies a session ID.
List → download	Find /workspace/outputs/agent.json. Retrieve its content and hash the saved bytes.

Commission a Bergen-to-Oslo desk

Call Studio in Inspector. Speak a four-tool travel-agent brief and listen for its confirmation.

1

First try: decline

Inspect the confirmation. Declining should leave the build unstarted.

2

Second try: approve once

If using a funded project, confirm one build. Keep the session ID and candidate receipt.

Run the agent Studio commissioned

Review the hosted candidate, approve to a new folder, then stop Studio.

```
Build

bash desk approve runs/YOUR_RUN_ID builds/studio-desk
```

Run

bash desk inspect builds/studio-desk

Produce a briefing someone can act on

Ask for tonight’s Oslo departures to Bergen, weather at landing and $200 in Norwegian kroner.

1

Choose the flight

Resolve date and route. Prefer estimated arrival; say when only the schedule is available.

2

Complete the briefing

Match the forecast hour to arrival. Quote the FX reference date. Mark unavailable data.

One flight number. Two possible forecasts.

Classroom fixture · ZZ201 from Oslo to Bergen. The traveler asks what to pack.

Date	Scheduled arrival	Estimated arrival
3 November	19:20	20:10
4 November	19:20	Unavailable

All times are local to Bergen. What must the desk ask before fetching the forecast?

Call the generated desk with the same two records

Stop the live Inspector. Start the approved four-tool project in rehearsal mode.

```
bash desk rehearse builds/studio-desk
```

Say “November third.”

Inspect the weather call. An answer of 8 °C points to the scheduled hour. The updated arrival points to 4 °C.

What will you ask the factory to build next?

Keep the run, hash, mode, first tool event and one failure case.

1

Change the experience

Commission a Spanish-speaking travel desk. Compare the generated skill and the tool calls.

2

Extend the tools

Ask Codex to add HND and a Tokyo-local-time test. Regenerate into a fresh directory.